

StratoWatch: Air Quality Forecasting with Spatio-Temporal Transformers and Graph-Aware Modeling

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Abstract—Forecasting ground-level ozone (O_3) and nitrogen dioxide (NO_2) a few hours ahead is difficult because pollutant concentrations depend on both recent atmospheric conditions and rapidly changing meteorological patterns. StratoWatch is a forecasting and evaluation framework motivated by ISRO Problem Statement 25178 that studies this problem at both single-site and multi-site scales. At the single-site level, we compare a forecast-only baseline with Random Forest, XGBoost, LSTM, TCN, and a temporal Transformer using a 24-hour input window and a 6-hour forecast horizon. XGBoost gives the strongest single-site result, with an MAE of 0.4039 and an R^2 of 0.3848 in the near-normalized evaluation units. We then extend the problem to seven Delhi monitoring stations using a spatio-temporal Transformer and two graph-based variants. The plain ST-Transformer obtains the lowest MAE at 18.8349 $\mu g/m^3$, while the Dynamic Wind Graph ST-Transformer obtains the lowest RMSE of 26.9455 $\mu g/m^3$ and the highest R^2 of 0.3911. These results suggest that explicit spatial modeling is useful, although its benefit depends on the evaluation metric rather than producing a uniform improvement. The current system is a short-horizon research prototype rather than a complete 24–48 hour operational forecasting system. Alongside the experiments, StratoWatch provides a React/FastAPI interface for running the forecasting pipelines and viewing their results.

Index Terms—air quality forecasting, ozone, nitrogen dioxide, transformers, spatio-temporal forecasting, graph neural networks, Delhi, ISRO

I. INTRODUCTION

Air-quality forecasting is difficult because pollutant concentrations are affected by several processes at the same time. O_3 and NO_2 respond to recent concentration history, wind, temperature, humidity, boundary-layer conditions, and other atmospheric variables. These effects can change considerably from one hour to the next. This makes short-term forecasting different from simply predicting a smooth time series: a model also has to deal with sudden changes and pollution peaks.

Physics-based forecasts provide an important starting point for this problem, but their predictions can still contain systematic errors at individual locations. Machine learning offers a way to learn patterns in historical observations and environmental variables that may not be fully captured by the initial forecast.

This work was motivated by ISRO Problem Statement 25178, which focuses on automated forecasting of surface-level O_3 and NO_2 using artificial intelligence and machine learning. The

implementation presented here should be viewed as a research prototype motivated by that problem rather than a complete implementation of its 24–48-hour operational requirement. The experiments currently use a six-hour forecast horizon so that the different model architectures can be compared under the same evaluation setting.

The study has two main parts. The first evaluates forecasting models at a single-site level. The second moves to a seven-site setting and asks whether explicitly representing relationships between monitoring stations improves forecasting performance.

The second question is particularly interesting because nearby stations are not independent. Pollution can move between locations, and wind can provide information about the direction of that movement. At the same time, meteorological variables already provide some of this information directly to a forecasting model. It is therefore not obvious how much additional information an explicit spatial graph will provide.

The main contributions of this work are:

- 1) A single-site forecasting pipeline comparing tree-based, recurrent, convolutional, and Transformer models for six-hour O_3 and NO_2 forecasting.
- 2) A seven-station spatio-temporal forecasting pipeline using a shared Transformer representation across monitoring stations.
- 3) Static and wind-aware dynamic graph variants for testing the effect of explicit spatial relationships between stations.
- 4) A reproducible evaluation structure based on frozen model artifacts and consolidated evaluation results.
- 5) A working web-based interface that connects the forecasting models to a FastAPI backend and allows results to be inspected without running the research code manually.

II. RELATED WORK

A. Machine Learning for Air Quality Forecasting

Machine learning has been widely used for air-quality forecasting because pollutant concentrations depend on nonlinear relationships between historical measurements and environmental variables. Tree-based methods are particularly useful when the input contains many engineered variables and nonlinear interactions. XGBoost provides an efficient gradient boosting framework for this type of tabular learning problem [2].

B. Sequence Modeling

Recurrent models such as Long Short-Term Memory (LSTM) networks are designed to capture dependencies in sequential data and have therefore been used extensively for environmental forecasting [3]. Temporal Convolutional Networks (TCNs) provide another approach, using causal and dilated convolutions to model dependencies over different temporal scales [4].

C. Transformers for Time-Series Forecasting

Transformers use self-attention to allow information from different positions in a sequence to interact directly [1]. This makes them attractive for forecasting problems where recent and older observations may both contain useful information. Informer further explored efficient Transformer-based forecasting for long sequences [6]. In StratoWatch, Transformer encoders are used to model temporal dependencies over a fixed 24-hour input window.

D. Spatio-Temporal Graph Models

When observations come from multiple monitoring locations, the spatial relationship between stations becomes another possible source of information. Graph-based forecasting methods represent locations as nodes and use edges to exchange information between them. Examples include Graph WaveNet, AGCRN, and DCRNN [5], [7], [8]. Recent air-quality work has also explored combinations of graph and Transformer architectures, including CEEMDAN-GNN-Transformer and extreme-event-oriented spatio-temporal graph models [9], [10].

These approaches motivate the multi-site experiments in StratoWatch, where spatial graph information is compared against a Transformer that already receives meteorological and multi-station information.

III. PROBLEM FORMULATION

For the single-site experiment, the model receives a historical window of observations and engineered environmental features. Let

$$X_t \in \mathbb{R}^{T \times F} \quad (1)$$

denote the input sequence, where $T = 24$ hours and $F = 134$ input features. The model predicts both pollutants over the next six hours:

$$\hat{Y}_t \in \mathbb{R}^{H \times C}, \quad (2)$$

where $H = 6$ and $C = 2$, corresponding to O_3 and NO_2 .

The final single-site XGBoost model uses direct target prediction rather than residual correction. The 24-hour history is flattened into a 3,216-dimensional feature vector (24×134), and twelve independent regressors are used for the six forecast horizons and two target pollutants.

For the multi-site experiment, the input is represented as

$$X \in \mathbb{R}^{T \times S \times F}, \quad (3)$$

where $T = 24$ hours, $S = 7$ monitoring stations, and $F = 28$ features per station. The output is

$$\hat{Y} \in \mathbb{R}^{H \times S \times C}, \quad (4)$$

where $H = 6$ forecast hours and $C = 2$ target pollutants.

IV. DATASET AND PREPROCESSING

A. Data Sources

The forecasting pipelines combine ground-level pollutant observations with meteorological, satellite-derived, and forecast-related variables. The data are aligned to an hourly time grid before being converted into model inputs.

TABLE I
MAIN DATA SOURCES USED BY STRATOWATCH

Source	Variables	Purpose
Ground stations Meteorology	Hourly O_3 , NO_2 Temperature, humidity, wind, boundary-layer variables	Target observations Atmospheric context
Satellite data	Trace-gas observations	Regional chemical context
Forecast variables	Forecast-related pollutant/environmental variables	Additional predictive information

Seven monitoring stations are used for the final multi-site experiment. The final single-site pipeline uses 134 input features, while the multi-site representation contains 28 features per station.

B. Preprocessing

The data are aligned to an hourly time grid and transformed into fixed-length historical windows. Feature scaling is performed using the appropriate training-derived statistics.

The feature construction includes temporal information and meteorological variables, with lagged and rolling information used where available. Wind direction is represented in a form that can be used by the forecasting models rather than treating the circular direction value as an ordinary scalar.

A chronological 70/15/15 train/validation/test split is used for the model experiments. The test data are kept separate from training and model-selection decisions.

C. Normalization and Evaluation Units

An important distinction exists between the two final experiments. The final single-site evaluation operates in near-normalized target units because the official single-site dataset is already normalized and the corresponding target scaler has mean values close to zero and unit scale. Its MAE and RMSE therefore should not be interpreted directly as $\mu\text{g}/\text{m}^3$.

The final multi-site results are inverse-scaled into real concentration units using the final target scaler. The target statistics are approximately $31.967 \mu\text{g}/\text{m}^3$ mean and $34.720 \mu\text{g}/\text{m}^3$ standard deviation for O_3 , and $39.133 \mu\text{g}/\text{m}^3$ mean and $29.909 \mu\text{g}/\text{m}^3$ standard deviation for NO_2 .

This distinction is important when comparing results across the two experiments.

D. Graph Construction

Two graph variants are evaluated in addition to the plain ST-Transformer.

The static graph represents fixed spatial relationships between stations using geographic proximity. Each station is connected to its nearest neighbouring stations using distance-based weights.

The dynamic wind graph retains candidate spatial relationships but changes their effective strength using the prevailing wind information. The purpose is to test whether a graph that changes with atmospheric transport conditions provides useful information beyond the meteorological variables already supplied to the model.

V. PROPOSED METHODOLOGY

A. Single-Site Models

Five final single-site model families are compared: Forecast-only Baseline, Random Forest, XGBoost, TCN, LSTM, and a temporal Transformer.

The Forecast-only Baseline provides a reference point without an additional learned forecasting model. Random Forest provides a non-neural tree ensemble, while XGBoost provides gradient boosted decision trees. The sequence models use the same general 24-hour forecasting setting but learn temporal representations differently.

The final XGBoost implementation is a direct-target model. It uses twelve independent XGBRegressor models, corresponding to each combination of six forecast horizons and two target pollutants. The input contains 24 hours of 134 features, resulting in 3,216 input values per sample.

B. Temporal Models

The LSTM processes the historical sequence recurrently, allowing information from earlier hours to influence the forecast. The TCN uses temporal convolutions to capture dependencies over different receptive-field sizes.

The temporal Transformer first projects the input features into a latent representation and then applies self-attention across the historical time dimension. The final model configuration uses a model dimension of 128, four attention heads, two temporal layers, dropout of 0.1, SmoothL1Loss, and AdamW optimization.

C. Multi-Site ST-Transformer

The multi-site ST-Transformer processes the seven monitoring stations jointly. Each station provides 28 features per hour over the 24-hour input window. Temporal dependencies are modeled before allowing information from the different stations to interact.

The model predicts the complete six-hour horizon directly rather than generating one step and feeding that prediction back into the model. This avoids the accumulation of autoregressive errors across the six forecast steps.

D. Graph-Enhanced Models

The Static Graph ST-Transformer adds fixed spatial information between stations. The Dynamic Wind Graph ST-Transformer extends this idea by changing graph connectivity strength according to wind conditions.

The purpose of these models is not to assume that a graph must improve performance. Instead, the experiments test whether explicit spatial structure provides useful information once multi-station observations and meteorological variables are already available.

TABLE II
MAIN MODEL CONFIGURATION

Parameter	Value
Input window	24 hours
Forecast horizon	6 hours
Multi-site stations	7
Single-site features	134
Multi-site features/site	28
Targets	O ₃ , NO ₂
Transformer dimension	128
Attention heads	4
Temporal layers	2
Dropout	0.1
Loss	SmoothL1Loss
Optimizer	AdamW
Early stopping	Patience 5
Random seed for Transformer models	42

VI. EXPERIMENTS AND EVALUATION

A. Experimental Setup

The final results are taken from frozen evaluation artifacts produced after the model experiments were completed. The research figures are generated from these consolidated results; no additional training is performed during figure generation.

The single-site test set contains 6,464 samples. The multi-site evaluation uses the frozen seven-station configuration and reports results after inverse scaling to real concentration units.

B. Evaluation Metrics

Three standard metrics are used:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (5)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (6)$$

and

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (7)$$

MAE measures the average absolute forecasting error, RMSE gives greater weight to larger errors, and R^2 measures the fraction of variance explained relative to the mean predictor.

VII. RESULTS AND DISCUSSION

A. Single-Site Results

Table III shows the final single-site comparison. All values are reported in the near-normalized evaluation units described in Section IV-C.

XGBoost gives the strongest overall single-site result across all three reported metrics. Its MAE is 0.4039, compared with 0.4392 for Random Forest. Its R^2 of 0.3848 also exceeds the other evaluated models.

The result is useful because the single-site input contains a relatively large number of engineered features compared with the length of the historical window. In this setting, the tree-based models appear to make more effective use of the tabular representation than the sequence models evaluated here.

The Forecast-only Baseline performs substantially worse, with an MAE of 1.2384 and a negative R^2 of -3.0296. This provides a useful reference for the learned models, although the near-normalized evaluation units mean that the magnitude should not be interpreted as a physical concentration error.

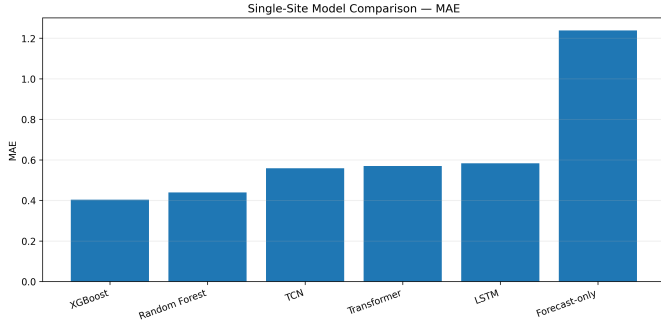


Fig. 1. Final single-site MAE comparison across the evaluated models.

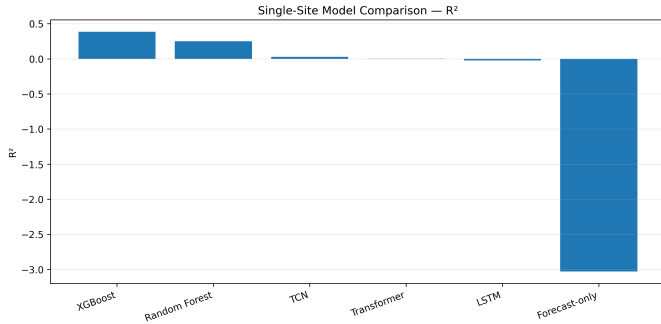


Fig. 2. Final single-site R^2 comparison across the evaluated models.

B. Single-Site Target and Horizon Analysis

For the final XGBoost model, the O_3 MAE is 0.3045 while the NO_2 MAE is 0.5034. The corresponding R^2 values are 0.0609 and 0.3561, respectively. This indicates that the model explains more of the variation in NO_2 than in O_3 under the current evaluation setup.

Performance also changes with forecast horizon. XGBoost obtains an MAE of 0.3497 at H+1, which gradually increases to

0.4451 at H+6. Its R^2 decreases from 0.5306 at H+1 to 0.2673 at H+6.

The gradual increase in error with horizon is expected in a short-term forecasting problem: as the prediction moves further away from the observed history, uncertainty increases. The result also shows that the first forecast hour contains considerably more predictable information than the later hours.

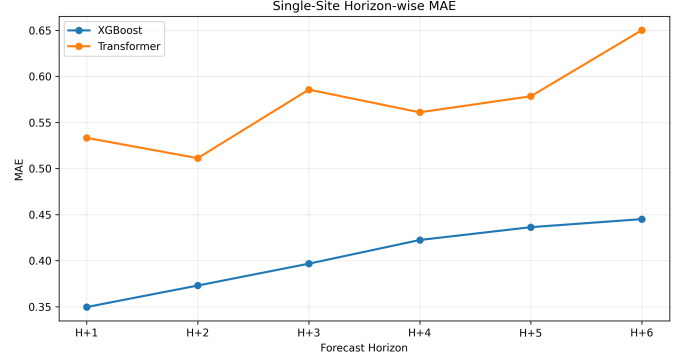


Fig. 3. XGBoost MAE across the six forecast horizons in the single-site experiment.

C. Multi-Site Results

The multi-site results are reported in real concentration units after inverse scaling. Table V compares the plain ST-Transformer with the two graph-enhanced variants.

The three models are relatively close in overall performance, but they do not produce the same ranking for every metric. The plain ST-Transformer obtains the lowest MAE at $18.8349 \mu g/m^3$. The Static Graph ST-Transformer improves on it in RMSE and R^2 , while the Dynamic Wind Graph ST-Transformer gives the best RMSE of $26.9455 \mu g/m^3$ and the highest R^2 of 0.3911.

This is different from simply saying that graph modeling either works or does not work. Under the current configuration, graph modeling provides a modest improvement in metrics that are more sensitive to larger errors and explained variance, while the plain ST-Transformer retains the best average absolute error.

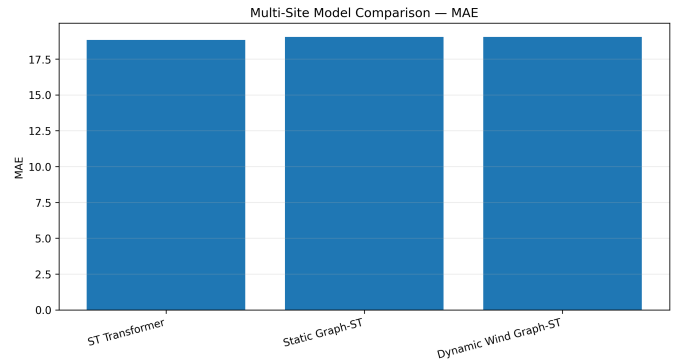


Fig. 4. Final multi-site MAE comparison.

TABLE III
FINAL SINGLE-SITE MODEL PERFORMANCE IN NEAR-NORMALIZED UNITS

Model	MAE	RMSE	R^2
Forecast-only Baseline	1.2384	1.5116	-3.0296
Random Forest	0.4392	0.6526	0.2489
XGBoost	0.4039	0.5906	0.3848
TCN	0.5588	0.7427	0.0273
LSTM	0.6616	0.8296	-0.2138
Transformer	0.5700	0.7524	0.0017

TABLE IV
FINAL XGBOOST HORIZON-WISE PERFORMANCE

Horizon	MAE	RMSE	R^2
H+1	0.3497	0.5157	0.5306
H+2	0.3731	0.5469	0.4723
H+3	0.3968	0.5810	0.4046
H+4	0.4225	0.6145	0.3342
H+5	0.4364	0.6303	0.2998
H+6	0.4451	0.6448	0.2673

TABLE V
FINAL MULTI-SITE MODEL PERFORMANCE IN $\mu\text{g}/\text{m}^3$

Model	MAE	RMSE	R^2
ST-Transformer	18.8349	27.9747	0.3437
Static Graph ST	19.0384	27.0781	0.3850
Dynamic Wind Graph ST	19.0464	26.9455	0.3911

D. Multi-Site Target Analysis

For the ST-Transformer, O_3 obtains an MAE of $19.0729 \mu\text{g}/\text{m}^3$ and an R^2 of 0.4078, while NO_2 obtains an MAE of $18.5969 \mu\text{g}/\text{m}^3$ and an R^2 of 0.2078.

For the Dynamic Wind Graph ST-Transformer, O_3 obtains an MAE of $18.7858 \mu\text{g}/\text{m}^3$ and an R^2 of 0.4769. NO_2 obtains an MAE of $19.3071 \mu\text{g}/\text{m}^3$ and an R^2 of 0.2226.

The difference between the pollutants is therefore not uniform across all metrics. The dynamic graph performs particularly well for O_3 in terms of explained variance, while its NO_2 MAE is slightly higher.

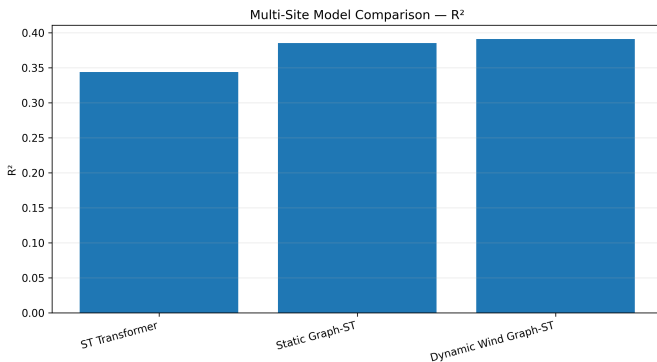


Fig. 5. Final multi-site R^2 comparison.

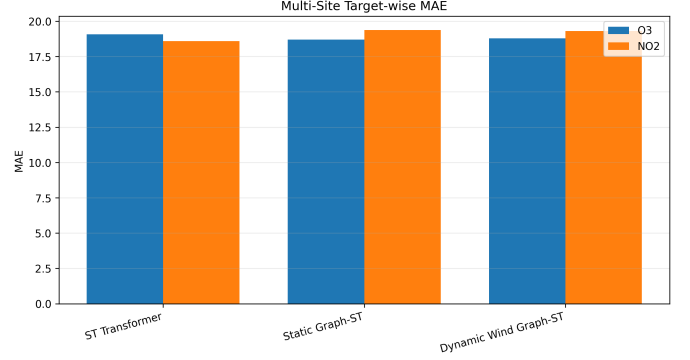


Fig. 6. O_3 and NO_2 MAE comparison for the final multi-site models.

E. Multi-Site Horizon Analysis

The Static Graph and Dynamic Wind Graph models show a gradual increase in MAE as the forecast moves from H+1 to H+6. For the Dynamic Wind Graph ST-Transformer, MAE increases from $18.8561 \mu\text{g}/\text{m}^3$ at H+1 to $19.2365 \mu\text{g}/\text{m}^3$ at H+6.

Its R^2 remains relatively stable over the six horizons, ranging from 0.3975 at H+1 to 0.3752 at H+6. This suggests that although the absolute error increases with horizon, the overall explained variance does not collapse over the six-hour prediction window.

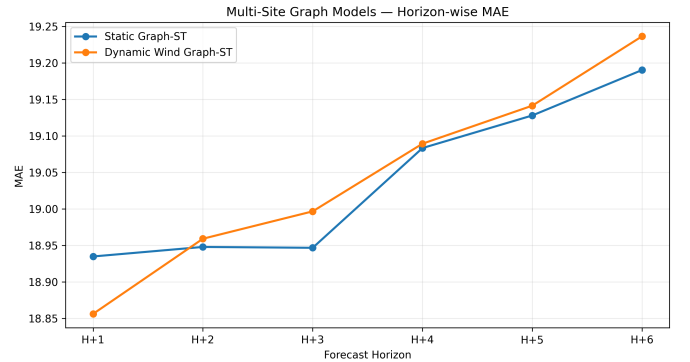


Fig. 7. Horizon-wise MAE for the final multi-site graph models.

VIII. QUALITATIVE ANALYSIS

Quantitative metrics describe average behaviour, but they do not show exactly where a model struggles. The final

evaluation therefore includes prediction-versus-actual plots for the pollutants.

Figure 8 shows the final O_3 prediction comparison, while Figure 9 shows the corresponding NO_2 result.

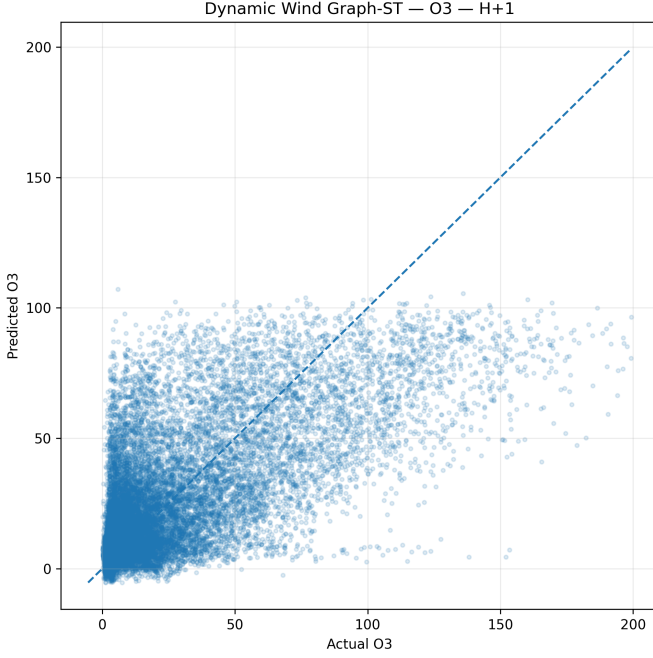


Fig. 8. Prediction versus actual O_3 values from the final evaluation.

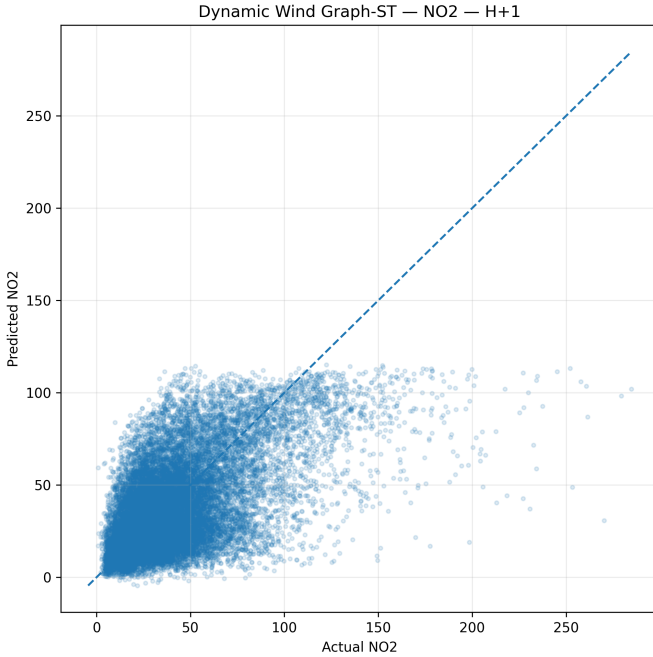


Fig. 9. Prediction versus actual NO_2 values from the final evaluation.

The final evaluation also shows a tendency to underestimate some high-pollution episodes. This behaviour matters because an early-warning system is particularly sensitive to missed

peaks. Average error metrics alone can hide this failure mode, especially when most samples are closer to normal concentration levels.

The present results therefore suggest that future work should evaluate models not only by overall MAE and RMSE, but also by their behaviour during high-concentration events.

IX. SYSTEM IMPLEMENTATION

The research models are integrated into the StratoWatch web application. The frontend is implemented using React and Vite, while the backend uses FastAPI. The backend exposes separate interfaces for the single-site and multi-site forecasting pipelines.

The application allows a user to provide input data, select the forecasting workflow, and inspect the resulting predictions, metrics, and visualizations. The final backend uses the frozen model artifacts rather than starting a new training process when a forecast request is made.

This separation is useful for the research workflow because the models used to produce the reported results can remain fixed while the web interface is used for inference and demonstration.

The deployed system is therefore not intended to replace the research evaluation pipeline. Instead, it provides a practical interface around the finalized forecasting models and makes the experimental work easier to inspect and demonstrate.

X. LIMITATIONS AND FUTURE WORK

The most important limitation of the current study is the six-hour forecast horizon. The problem motivating the work targets substantially longer operational forecasts, including 24–48 hours. The current results should therefore not be interpreted as evidence that the system already satisfies that operational requirement.

A second limitation is the difference in evaluation units between the two experiments. Single-site metrics are reported in near-normalized units, while the multi-site metrics are reported in real concentration units. This prevents direct numerical comparison of the MAE values between the two experiments.

The multi-site study is also limited to seven monitoring stations. The final graph checkpoints were trained and evaluated for this configuration, so conclusions about other station layouts or larger geographic areas require additional experiments.

Another limitation is the difficulty of forecasting high-pollution spikes. The current models can underestimate extreme values, which is particularly important for an early-warning application. The present study does not include a dedicated extreme-event objective or uncertainty calibration.

Finally, the dynamic graph is based on available wind information and does not model the full vertical structure of atmospheric transport. Real pollutant movement can depend on processes that are not captured by a surface-wind-based relationship alone.

Future work will focus on:

- 1) Extending the forecasting horizon to 24 and 48 hours.
- 2) Improving the prediction of high-pollution episodes using spike-aware objectives or sampling strategies.
- 3) Adding calibrated predictive uncertainty to the forecasts.

- 4) Learning spatial relationships directly from data rather than relying only on predefined geographic and wind-based relationships.
- 5) Incorporating additional emission-related information such as traffic, fire, and industrial activity proxies where suitable data are available.
- 6) Connecting the deployed system to continuously updated meteorological and satellite feeds for genuinely real-time forecasting.

XI. CONCLUSION

StratoWatch presents a practical study of short-horizon O_3 and NO_2 forecasting at both single-site and multi-site scales. At the single-site level, XGBoost gives the strongest overall result among the evaluated models, with an MAE of 0.4039, RMSE of 0.5906, and R^2 of 0.3848 in the near-normalized evaluation units.

The multi-site experiment gives a more mixed picture. The plain ST-Transformer obtains the lowest MAE at $18.8349 \mu g/m^3$, while the Dynamic Wind Graph ST-Transformer obtains the lowest RMSE and highest R^2 . The Static Graph ST-Transformer remains competitive. These results suggest that explicit spatial relationships can provide useful information, but the benefit is metric-dependent rather than a uniform improvement over the non-graph model.

The experiments also show that forecasting becomes more difficult as the prediction horizon increases and that high-pollution episodes remain challenging. These observations are important because a forecasting system intended for early warning should eventually be evaluated specifically on the events that matter most operationally.

The current StratoWatch system is therefore best viewed as a short-horizon research prototype and evaluation platform. Its main value at this stage is not claiming a finished operational forecasting solution, but providing a reproducible comparison of several forecasting approaches and a working system in which those models can be evaluated and demonstrated.

DATA AND CODE AVAILABILITY

The StratoWatch source code, model implementations, pre-processing pipeline, evaluation artifacts, and web application are maintained in the project’s public GitHub repository.

Ground-level air-quality observations were obtained from CPCB-related sources, while satellite-derived atmospheric information is based on Sentinel-5P/TROPOMI products and other environmental data used by the project. Redistribution of third-party data remains subject to the terms and conditions of the respective providers.

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